

Xiaoling Sundberg

STA 6709

Semester Project

April 03, 2026

Executive Summary

This project applies spatial statistics and mapping methods to a Colorado Zillow housing dataset to examine how residential listing prices vary across geography and property characteristics. The analysis uses listing price, home type, city, ZIP code, beds, baths, square footage, latitude, and longitude to describe housing patterns across Colorado. Because each listing includes geographic coordinates, the homes were treated as spatial points and mapped within the Colorado state boundary and county polygons.

The cleaned dataset contains 123 Zillow listings across 44 cities and 82 ZIP codes. The median listing price was \$635,000, while the mean listing price was approximately \$3.97 million, showing that the price distribution is strongly right-skewed due to a small number of luxury listings. Most listings were single-family homes, with fewer condos and townhouses. Maps showed that listings were not evenly distributed across Colorado; instead, they were concentrated mainly along the Front Range, especially near Denver, Colorado Springs, Boulder, and nearby communities.

Regression analysis was used to examine how listing price relates to property characteristics and location. Because raw prices were highly skewed, log listing price was used as the response variable. The regression results showed that square footage, bathrooms, home type, days on Zillow, latitude, and longitude were meaningful predictors of listing price. The residual map also showed that some geographic patterns remained after modeling, suggesting that location-based factors not fully captured in the dataset may still affect housing prices.

Spatial point pattern analysis further showed that Zillow listings were clustered rather than randomly distributed. The nearest-neighbor index was 0.648, indicating that listings were closer together than expected under complete spatial randomness. A cluster analysis identified major listing groups along the Front Range, with the largest cluster around Denver and central Colorado. Overall, the project shows that spatial analysis provides valuable insight into Colorado housing patterns by combining summary statistics, regression modeling, static maps, and an interactive Leaflet map.